Are My Home Network Devices Really Spying on Me?

Securing your home network — and why most people don't bother

By Michael Coar

Smart TVs, speakers, appliances and even light bulbs are computers on your network. Most of us never see what they are doing. I decided to look.

When I started down the rabbit hole of home network security, I wondered whether I was just being paranoid. Then I looked at the logs. It is surprisingly difficult to remain relaxed when a smart light bulb appears to have a more active social life than you do.

Smart devices can collect data, contact cloud services and introduce security risks. But a busy network is not, by itself, proof of spying. Some activity is useful, some is advertising or analytics, and some is simply bad software. The challenge is figuring out which is which — and deciding how much effort you want to spend keeping your appliances in line.

Once I started looking at what was actually happening on my network, I discovered that some of my paranoia was apparently justified. And the journey started with something pretty mundane: ads.

Why Does the Internet Suddenly Know I Want a Dishwasher?

Search for a dishwasher and suddenly the Internet seems convinced that replacing your dishwasher is your entire personality. So, you install a browser ad blocker. Then half the websites complain about it and ask you to turn it off.

Browser ad blockers typically use maintained filter lists to block matching requests and hide advertising elements on a page. Some rules match domains, others match URL patterns, scripts or page elements. They are considerably more sophisticated than simply searching for the word "advertisement." [1]

They work quite well, but they protect the browser in which they are installed. Your smart TV does not care that Chrome has an ad blocker. Neither does Alexa, your smart light bulb, security camera, thermostat, washing machine or whatever other supposedly "smart" gadget you have invited onto your network.

To reach those devices, you can put the filter at another point in the process: DNS.

DNS: The Internet's Phone Book

DNS stands for Domain Name System. Its basic job is simple: it translates names humans understand into addresses computers understand.

When you type CNN.com, your computer needs the numerical IP address of the server hosting CNN. So, it asks a DNS server, essentially:

"Hey, where's CNN.com?"

The DNS server provides the IP address, and your browser uses that address to connect to the site.

For many households, DNS is supplied automatically by the Internet service provider. Traditional DNS queries are commonly unencrypted, so the resolver — and parties able to observe that traffic — can learn which domains are being requested. Encrypted DNS protocols such as DNS-over-HTTPS and DNS-over-TLS reduce that exposure, although they do not make Internet activity anonymous.[2][3]

Moving the Ad Blocker to the Entire Network

I eventually discovered Pi-hole, which takes a different approach. Instead of installing an ad blocker in every browser, you make Pi-hole part of the DNS path for your home network. It maintains lists of known advertising, tracking and other undesirable domains and can prevent devices across the network from resolving them.[4]

I installed Pi-hole on a Raspberry Pi and was surprised by the results. Using fairly ordinary blocking lists, roughly 25% of the DNS requests on my network were being blocked. That figure is specific to my network, but it got my attention. Who — or what — was making all of those requests?

Then I discovered Technitium DNS Server.

Technitium: This Is Where the Rabbit Hole Gets Deeper

Technitium can perform network-wide domain blocking like Pi-hole, but it is also a full DNS server. It can operate recursively or use upstream forwarders, cache DNS results, use encrypted DNS protocols, apply block lists, and provide query logging and statistics.[5]

In my setup, devices use Technitium for DNS rather than independently sending ordinary DNS queries to an ISP-provided resolver. Technitium can then cache answers and use encrypted DNS when communicating with an upstream resolver. That does not make the traffic anonymous, but it gives me far more control over where DNS requests go and much better visibility into what is requesting them.

The logging and statistics are what really opened my eyes. I could finally see what the devices in my house were actually doing.

My Light Bulb Would Not Shut Up

Shortly after installing Technitium, I noticed something astonishing: about 67% of the DNS requests on my network were failing. Not being blocked — failing. And they were happening continuously, 24 hours a day.

I eventually traced most of them to a single Sengled smart light bulb. The bulb itself still worked through Alexa, but it repeatedly tried to contact a server that was no longer responding. Sengled's cloud services have suffered prolonged outages and reports of discontinued support, although I could not find a definitive company shutdown notice. Whatever the corporate story, the practical result was the same: one forgotten light bulb was generating an extraordinary amount of useless network chatter.[6]

That illustrates a larger problem with Internet-connected devices. You are relying on the vendor to maintain firmware, address security problems and keep whatever cloud infrastructure the product depends on alive. An abandoned device that keeps trying to reach old infrastructure is at minimum noisy and, depending on how it was designed and secured, worth investigating.

Geesh. Something else to keep track of.

Unfortunately, None of This Is Particularly Easy

Installing something like Technitium is not terribly difficult if you already know how to do this stuff. For everyone else, any instructions that begin with "First, get a Raspberry Pi…" have probably already lost half the audience.

Then come operating systems, storage, static IP addresses, router DNS settings and administration. And you have to maintain it. Software needs updating. Block lists occasionally break things. Some websites refuse to behave when advertising or tracking systems are blocked.

Eventually someone in the house says:

"Can you just turn the damn thing off?"

Now we are beginning to understand why most people do not bother.

So, Am I Actually Being Paranoid?

Maybe. I have a friend who swears that after he and his wife discussed buying a toaster oven in their kitchen (where they have an Alexa device) — without ever searching for one online — toaster-oven ads began appearing whenever they opened a browser. Does that prove Alexa secretly listened and sold the information? No. Advertising systems correlate information in complicated ways, and anecdotes are not evidence. But I understand why the experience creeped him out.

At the same time, another friend who works for Amazon once told me that when executives rented a house for meetings, one of the first things they did was walk around and unplug the Alexa-type devices.

Again, anecdote, not proof. The more useful question is what smart devices are demonstrably capable of collecting.

But then there's this story: A friend's parents have a "smart" photo picture display in their living room, where family members are given access to it, so they can upload photos and videos to it. He came home one day, and found a message on the display saying something to the effect of: "This has been hacked. Its firmware is not safe, and you should throw this out." And he did throw it out.

Your Television Is a Computer Hanging on the Wall

A smart TV is not merely a television. It is a network-connected computer with a giant screen. The same observation applies to connected refrigerators, doorbells, light bulbs, outlets, thermostats and coffee makers.

There is strong precedent for privacy concerns. In 2017, the U.S. Federal Trade Commission and New Jersey settled charges alleging that Vizio collected second-by-second viewing information from millions of smart TVs without adequate disclosure or consent. Modern smart TVs also commonly use Automatic Content Recognition, or ACR, to identify what is being displayed and support advertising and analytics.[7][8]

More recently, independent testing reported that certain LG televisions scanned their local networks and identified other devices. LG acknowledges local-network discovery for connectivity and smart-home functions. Researchers also raised concerns about voice processing and locally retained data; LG disputes claims that its TVs continuously record or transmit ambient conversations, saying voice processing occurs only after an enabled wake word or voice-button activation. Network scanning is documented; the broader interpretation of the voice behavior remains disputed.[9][10]

There have also been reports of connected appliances generating unexpectedly large amounts of network traffic. One LG washing-machine owner reported about 3.66 GB in a day, although the cause was not established. In another case, a Keurig smart coffee maker reportedly generated roughly 1 TB of traffic over ten days, with reporting indicating that most of it apparently remained inside the local network. Fascinating? Absolutely. Proof that the manufacturers were exfiltrating that volume of personal data? No. [11][12]

Now Imagine the TV Is in a Boardroom

This is where the problem becomes more serious than targeted advertising. Imagine an Internet-connected television hanging on the wall of a corporate boardroom while people discuss acquisitions, unreleased products, financial results, patents or intellectual property.

Even if the manufacturer has no malicious intent, the television is another computer on the network. If it is compromised, poorly secured or running obsolete software, it becomes another potential attack surface.

Even the Router Matters

The router is arguably the most security-sensitive device in the house because it controls traffic between your network and the Internet. Router models from several major manufacturers have been exploited in malware and state-sponsored campaigns. That does not mean an entire brand is unsafe; it means buyers should favor supported hardware, install security updates, replace end-of-life equipment and disable unnecessary remote administration.[13][14]

When was the last time you even thought about your router? Many ISP-provided routers receive updates automatically. If you provide your own, you need to know whether it is still supported — and whether updates are actually being installed.

What Can You Actually Do About It?

You can build a substantially more secure home network. Doing it properly requires some money, time, technical knowledge — or hiring someone you trust to do it for you — and an acceptance that occasionally your wonderful security system is going to annoy you.

1. Separate Your Devices Into Different Networks

One of the most useful things you can do is use VLANs — Virtual Local Area Networks (also referred to as subnets) — to create separate logical neighborhoods inside your home network.

Trusted computers, laptops, tablets and phones can live on a main network. Smart TVs, Alexa devices, outlets, bulbs and appliances can live on an IoT network. Visitors can use a guest network. Firewall rules can then control which networks are allowed to initiate connections to the others.

If some future television decides it wants to inventory everything around it, I would prefer that it discover a rather boring neighborhood consisting mostly of itself and some light bulbs.

2. Use Separate WiFi Networks

Those VLANs can be associated with different WiFi names, or SSIDs: MAIN for trusted devices, CLOUDSTUFF for IoT gear, and GUEST for everyone else. When I install a new smart TV, I do not simply hand it the password to my primary WiFi network.

Welcome to CLOUDSTUFF. Stay on your side of the fence.

3. Use Managed Ethernet Switches

A managed switch can assign individual Ethernet ports to different VLANs. A desktop computer and a smart TV can be plugged into the same physical switch while remaining in different logical networks.

4. Consider Separate Wireless Access Points

Dedicated wireless access points can provide better placement and coverage while supporting multiple SSIDs associated with different VLANs. Wired Ethernet backhauls (where the WAP is plugged directly into your Ethernet network) is preferable when practical, although mesh links (where WAPs talk to each other, but require at least 1 WAP be connected directly to your network) can be useful where cabling is difficult.

5. Use Wired Ethernet Wherever Practical

Wired Ethernet is generally faster and more consistent than WiFi and avoids the radio interference inherent to wireless networking.

If your house lacks Ethernet but has coaxial cable from the cable-TV era, MoCA can repurpose that wiring for networking. MoCA 2.5 supports aggregate data rates up to 2.5 Gbps, and MoCA networking can coexist with compatible television signals on the same coax.[15]

6. Add Network-Level DNS Filtering

I use Technitium for DNS and network-level blocking. It gives me block lists, encrypted upstream DNS options, caching, logging and — most valuable to me — visibility into DNS activity.[5]

You can't investigate network behavior you don't know is occurring.

The Maintenance Is Part of the Purchase

The awkward part is that someone has to maintain all this. Blocking occasionally breaks things. I have encountered videos that would not play until I relaxed filtering. A narrow allowlist exception is preferable to switching off protection for everyone, and a temporary pause can help diagnose a problem. Neither approach is much fun when somebody just wants to watch the news.

Routers need attention too. The FBI has specifically warned about criminals exploiting end-of-life routers and recommends replacing unsupported equipment and limiting remote administration.[14]

In my house, updating more than twenty devices can consume a substantial part of a day. Automatic updates help where available, but a monthly reminder is not a reason to postpone an urgent security fix. Adding a device also means deciding where it belongs — and keeping the trusted WiFi password out of guest circulation.

So, Is All of This Worth It?

You do not have to begin by rebuilding your entire network. Start by checking whether your router is still supported, updating devices, reviewing privacy settings and using an isolated guest or smart-device network where available. Add DNS filtering if you are willing to handle the occasional exception. Move to VLANs and managed equipment when your needs justify the work.

For me, the visibility and control are worth the effort. I no longer assume every failed lookup is an attack, or every advertisement is evidence of a hidden microphone. But I also no longer assume a device is harmless because its main job is turning on a lamp.

A properly segmented home network can involve a capable router, VLANs, several WiFi networks, managed switches, access points, DNS filtering, firewall rules, troubleshooting and regular software updates.

That is a ridiculous amount of technology just so I can safely turn a light bulb on from my phone.

And that is exactly why most people do not do it.

Once everything is configured, there is not much day-to-day work. I update things periodically, deal with the occasional blocked service and think about which network a new device should join before connecting it.

Before I started this project, I could not see what the devices in my house were doing. Now I can. I have watched a single light bulb generate an astonishing number of failed DNS requests. I have seen how much advertising and tracking traffic network-level filtering stops. And I have learned just how many computers we casually bring into our homes disguised as televisions, appliances, speakers and light bulbs.

I would still prefer a home where the smartest thing about the coffee maker is the coffee. Until then, I want to know who my appliances are talking to — and whether they really need to be talking at all.

So perhaps the better question is not “Am I paranoid?”

“Why does my light bulb need to talk to the Internet at 3:00 in the morning?”

And if you do not know the answer, perhaps a little paranoia is not such a bad thing.

Sources & Notes

[1] Adblock Plus, “How to Write Filters” — request blocking, exceptions, element hiding and domain-specific filtering rules.

[2] Cloudflare, “DNS Encryption Explained” — background on traditional DNS and encrypted DNS protocols such as DoH and DoT.

[3] U.S. Federal Trade Commission, 2021 ISP privacy staff report — data collection by major Internet service providers and privacy implications of browsing/application data.

[4] Pi-hole documentation — network-wide DNS sinkhole that blocks unwanted domains without client-side software.

[5] Technitium DNS Server official documentation — recursive/forwarding DNS, block lists, DoT/DoH/DoQ, caching, query logging and statistics.

[6] Public reports and customer complaints concerning Sengled cloud-service outages and discontinued support during 2025–2026. No definitive corporate shutdown notice was located; the article therefore describes the observed service failure rather than asserting a formal company closure.

[7] U.S. Federal Trade Commission, 2017 Vizio settlement — alleged collection of second-by-second viewing information from approximately 11 million smart TVs without adequate disclosure or consent.

[8] Consumer Reports guidance on smart-TV Automatic Content Recognition (ACR) — viewing recognition and privacy controls.

[9] Ars Technica, September 2026, reporting on independent testing of LG televisions — local-network scanning/device discovery and researchers’ voice-data findings.

[10] LG Electronics, September 2026, “Understanding Privacy on LG Smart TVs” — confirms device-discovery functions and disputes continuous recording/transmission of ambient conversations.

[11] Tom’s Hardware, 2024 — report concerning an LG washing machine showing approximately 3.66 GB of traffic in one day; cause not established.

[12] Tom’s Hardware, October 2026 — report concerning a Keurig smart coffee maker generating roughly 1 TB of traffic over ten days; most traffic reportedly remained on the local network.

[13] U.S. Department of Justice, April 2026 — court-authorized disruption involving Russian GRU exploitation of known vulnerabilities in thousands of TP-Link routers; D-Link and CISA advisories document other exploited networking equipment.

[14] Federal Bureau of Investigation, May 7, 2025, “Cybercriminal Proxy Services Exploiting End-of-Life Routers” — recommends replacing unsupported equipment and limiting remote administration.

[15] Multimedia over Coax Alliance, MoCA 2.5 specification — up to 2.5 Gbps aggregate throughput over coax.

Publication Note

Personal observations in this article — including the percentages of blocked or failed DNS requests on the author’s own network and the Sengled light-bulb incident — are firsthand observations, not controlled measurements and should not be interpreted as representative of other home networks.